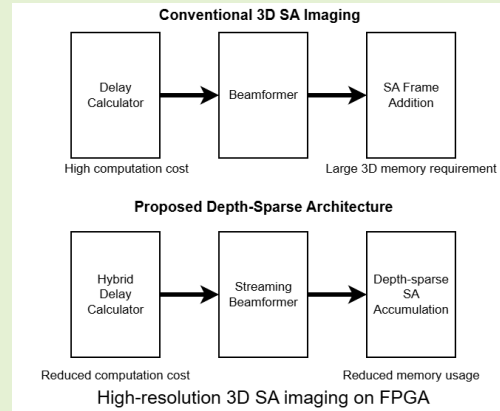


Depth-Sparse 3-D Synthetic Aperture Ultrasonic Imaging with a Fully On-Chip Beamforming Architecture

Yu-Xiang Hung, Po-Jui Su, Geng-Shi Jeng, and Bo-Cheng Lai

Abstract—Three-dimensional environment sensing is increasingly critical for autonomous vehicles, drones, and smart manufacturing systems. Conventional modalities, including radar, light detection and ranging, and visible-light cameras, each have limitations in angular resolution, cost, weather robustness, or direct depth sensing. Air-coupled ultrasound offers a low-cost and weather-resilient alternative, but conventional ultrasonic systems are often restricted to coarse ranging or low-resolution directional estimation. Extending synthetic aperture imaging to three dimensions further introduces substantial computational and memory demands for fully on-chip field-programmable gate array implementation. This paper presents a depth-sparse 3-D synthetic aperture ultrasonic imaging architecture with a fully on-chip beamforming pipeline implemented on an AMD/Xilinx ZCU102 field-programmable gate array. The proposed design integrates hybrid delay generation, which combines exact transmit-distance computation with lookup-table-assisted linear receive-delay approximation, a channel-parallel streaming beamformer with coherence factor weighting, and a depth-sparse synthetic aperture accumulation architecture. By retaining only the K most significant voxel responses per scanline, the design achieves an $8.70\times$ memory compression ratio and an 88.51% memory reduction in on-chip storage, enabling fully on-chip multi-transmit synthetic aperture reconstruction without external memory access. Experimental results within a $\pm 60^\circ$ field of view and a 0.3–2.0 m sensing range demonstrate a frame rate of 68.56 frames/s and a best angular resolution of 2.3° at 0.5 m.

Index Terms—Air-coupled ultrasound, field-programmable gate array (FPGA), real-time 3-D sensing, synthetic aperture imaging, ultrasonic beamforming.



I. INTRODUCTION

THREE-DIMENSIONAL (3-D) environment sensing is essential for autonomous vehicles, drones, smart factories, and warehouse automation. Existing sensing modalities, such as LiDAR, radar, and visible-light cameras, provide useful perception capabilities but also have practical limitations in cost, angular resolution, weather robustness, or direct depth acquisition [1]–[5]. Air-coupled ultrasound is therefore an attractive complementary modality because of its low cost and robustness in scenarios where optical sensing may degrade [6], [7]. However, conventional ultrasonic sensing systems are often limited to one-dimensional ranging or coarse directional estimation, which restricts their use in high-resolution 3-D perception.

This paragraph of the first footnote will contain the date on which the paper was submitted for review and the financial support acknowledgment, if applicable.

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Synthetic aperture (SA) imaging can improve angular resolution by coherently combining multiple transmit–receive events, and has been widely studied in medical ultrasound and sonar systems [8]–[10]. Nevertheless, directly extending SA imaging to 3-D air-coupled ultrasound introduces substantial computational and memory overhead. In particular, voxel-wise delay generation requires intensive arithmetic operations, while multi-transmit SA reconstruction requires storing and accumulating partial volumetric images. This dense intermediate buffering becomes a major obstacle for fully on-chip FPGA implementation, especially under a wide field of view and a large 3-D imaging volume.

To address these challenges, this work proposes a *depth-sparse 3-D synthetic aperture ultrasonic imaging architecture with fully on-chip beamforming*. The proposed design combines hybrid delay generation, a channel-parallel streaming beamformer with coherence-factor weighting, and a depth-sparse SA accumulation architecture. Instead of storing the full 3-D partial image volume, only the K most significant voxel responses along each scanline are retained, enabling multi-transmit SA reconstruction to be performed entirely on chip.

The proposed architecture is implemented on an

AMD/Xilinx ZCU102 FPGA platform for 40-kHz air-coupled ultrasonic imaging. Under a $\pm 60^\circ$ field of view and a sensing range of 0.3–2.0 m, the system achieves a frame rate of 68.56 frames/s and a best angular resolution of 2.3° . The proposed sparse representation further achieves an $8.70\times$ memory compression ratio and an 88.51% memory reduction in on-chip storage, demonstrating that high-resolution 3-D SA imaging can be realized with practical FPGA resource usage.

The main contributions of this work are summarized as follows:

- A depth-sparse 3-D SA reconstruction flow is proposed to avoid dense partial-volume buffering for fully on-chip FPGA implementation.
- A compact sparse memory format with streaming Top- K update is developed, reducing intermediate storage by 88.51%.
- A fully streaming FPGA beamforming architecture integrating hybrid delay generation, DAS/CF computation, and SA accumulation is implemented on ZCU102.
- Real-time 3-D air-coupled SA imaging is demonstrated with 68.56 frames/s and a best angular resolution of 2.3° over a $\pm 60^\circ$ field of view.

The remainder of this paper is organized as follows. Section II reviews related work on air-coupled ultrasonic imaging and FPGA-based beamforming. Section III presents the proposed architecture. Section IV describes the experimental setup and evaluation methodology. Section V reports the imaging and hardware results. Section VI concludes this paper.

II. BACKGROUND AND RELATED WORK

A. Air-Coupled Ultrasonic Imaging and Challenges

Air-coupled ultrasound is a promising sensing modality for environment perception because of its low cost, robustness to optical degradation, and suitability for short- to mid-range sensing. However, compared with medical ultrasound, air-coupled imaging suffers from lower signal-to-noise ratio (SNR), stronger attenuation [11], [12], and a larger region of interest, which together make high-resolution 3-D imaging more difficult. In addition, conventional ultrasonic sensing systems are often limited to simple ranging or low-angular-resolution detection due to the restricted physical aperture and channel count.

B. Synthetic Aperture Imaging for Resolution Enhancement

Synthetic aperture (SA) imaging has been widely studied in medical ultrasound [8], [9] and sonar systems [10] as an effective approach for improving spatial resolution and image quality. By coherently combining echoes acquired from multiple transmit–receive events, SA imaging synthesizes a larger effective aperture and achieves better focusing performance than conventional delay-and-sum beamforming. Similar concepts have also been applied to air-coupled and airborne ultrasound imaging.

However, directly extending SA imaging to 3-D air-coupled ultrasound introduces substantial computational and memory overhead, since voxel responses from multiple transmit events must be reconstructed and accumulated. In particular, dense SA reconstruction requires buffering multiple partial volumetric images or repeatedly accessing intermediate voxel responses, which is impractical for fully on-chip FPGA implementation under a large field of view and sensing range. This memory bottleneck motivates the proposed depth-sparse SA accumulation strategy, which preserves only significant depth responses along each scanline and performs multi-transmit accumulation using a compact sparse representation.

C. FPGA-Based Beamforming and On-Chip Implementation Challenges

FPGA-based acceleration is attractive for real-time ultrasonic imaging because it offers high throughput and energy-efficient signal processing [13], [14]. Nevertheless, fully on-chip implementation of 3-D SA imaging remains challenging. Two major bottlenecks are the large amount of delay calculation required for voxel-wise beamforming and the memory overhead caused by storing and accumulating multiple partial images across transmit events.

Although prior systems have demonstrated real-time ultrasonic imaging [6], [7], they often do so by sacrificing angular resolution or limiting reconstruction complexity in order to meet latency and hardware constraints. In contrast, high-resolution 3-D synthetic aperture imaging requires the reconstruction and accumulation of dense voxel responses over multiple transmit events, leading to substantial computational overhead, memory demand, and processing latency. This resolution–latency tradeoff motivates the proposed architecture, which improves both computational and memory efficiency through delay approximation, streaming beamforming, and depth-sparse accumulation.

III. PROPOSED ARCHITECTURE

This section presents the proposed FPGA architecture for real-time 3-D synthetic aperture (SA) ultrasonic imaging. The design addresses two major bottlenecks in fully on-chip 3-D SA reconstruction: the high arithmetic cost of voxel-wise delay generation and the large memory overhead of storing multi-transmit partial images. To overcome these challenges, the proposed architecture integrates hybrid delay generation, streaming DAS/CF beamforming, and depth-sparse SA accumulation into a unified on-chip dataflow.

A. System Overview

Fig. 1 shows the overall architecture of the proposed system. For each transmit event, the delay calculator generates the channel-wise sample indices and phase references required for voxel-wise beamforming. The delayed RF samples are then fetched from the input buffer and processed by the streaming beamformer, which performs quadrature demodulation, delay-and-sum (DAS) accumulation, and coherence-factor (CF) weighting.

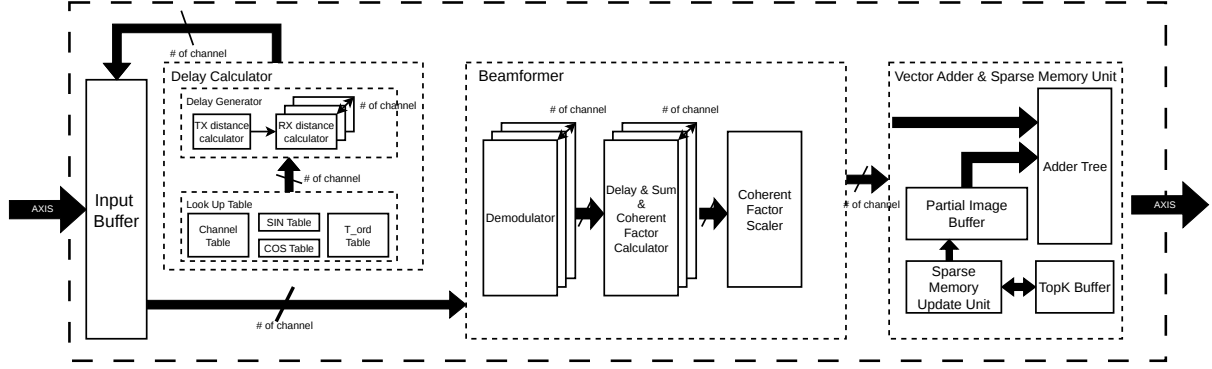


Fig. 1. Overall architecture of the proposed depth-sparse 3-D synthetic aperture ultrasonic imaging system. The system consists of a hybrid delay calculator, a streaming DAS/CF beamformer, and a depth-sparse SA accumulation unit for fully on-chip reconstruction.

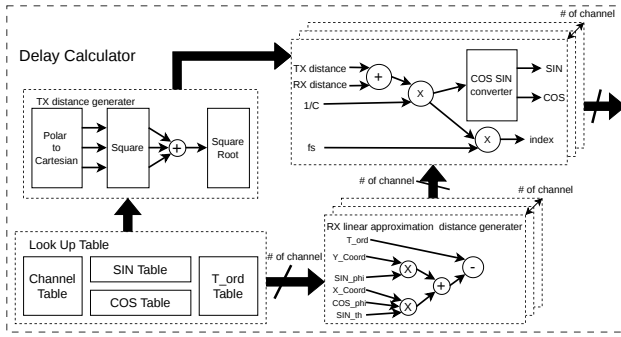


Fig. 2. Architecture of the proposed hybrid delay calculator. The transmit distance is computed exactly, while the receive delay is approximated using a lookup-table-assisted linear model.

The resulting voxel stream is forwarded to the depth-sparse SA accumulation unit. Instead of buffering dense partial 3-D images for all transmit events, the proposed architecture stores only the significant depth responses along each scanline. The sparse responses from multiple transmit events are then accumulated on chip to form the final SA image. This dataflow avoids external memory access for intermediate reconstruction and enables real-time multi-transmit 3-D SA imaging on FPGA.

B. Hybrid Delay Generation

Fig. 2 illustrates the proposed hybrid delay calculator. Since the transmit distance is shared by all receive channels for a given voxel and transmit event, it is computed exactly as

$$d_{TX} = \sqrt{(x_S - x_T)^2 + (y_S - y_T)^2 + (z_S - z_T)^2}, \quad (1)$$

where (x_S, y_S, z_S) denotes the voxel position and (x_T, y_T, z_T) denotes the active transmit element position.

In contrast, directly computing the receive distance for every receive channel would require repeated square-root operations and become expensive under channel-parallel processing. Therefore, the receive path adopts a first-order approximation supported by lookup tables:

$$t_{RX} \approx T_{ord} - \frac{x_D \sin \theta}{c} - \frac{y_D \sin \phi \cos \theta}{c}, \quad (2)$$

TABLE I

APPROXIMATION ERROR OF THE PROPOSED HARDWARE-ORIENTED RX DELAY CALCULATION.

Metric	Mean	RMSE	Max.
Distance error (mm)	0.216	0.401	5.195
Sample error (samples)	0.105	0.194	2.513
Phase error ($^\circ$)	9.167	17.001	220.024

where (x_D, y_D) denotes the receive-element coordinates, θ and ϕ are the steering angles, c is the speed of sound, and T_{ord} is the precomputed reference distance term.

The transmit and receive terms are combined to obtain the round-trip delay:

$$t_d = \frac{d_{TX}}{c} + t_{RX}, \quad (3)$$

which is then converted into the nearest lower RF sample index:

$$n = \lfloor t_d f_s \rfloor. \quad (4)$$

The fractional part of the delay is handled by the phase-reference rotation used in the demodulation path. The same delay value is also used to access the sine and cosine phase references for quadrature demodulation. By combining exact transmit-distance computation with approximation-based receive-delay generation, the proposed design removes per-channel square-root operations while sustaining channel-parallel streaming.

As summarized in Table I, the proposed approximation achieves a mean sample-index error of 0.105 samples and an RMSE of 0.194 samples. The maximum error occurs only under the worst-case geometry, corresponding to the nearest depth, the largest steering angle, and an edge receive channel.

C. Streaming DAS/CF Beamformer

Fig. 3 shows the proposed streaming beamformer. After the delayed RF samples are fetched, each channel is demodulated into in-phase and quadrature components using the phase references generated by the delay calculator:

$$I_m = r_m \cos \phi_m + q_m \sin \phi_m, \quad (5)$$

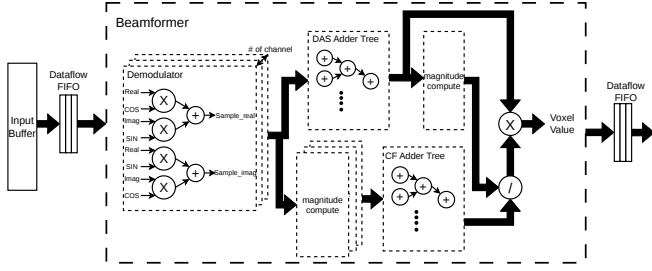


Fig. 3. Architecture of the proposed streaming DAS/CF beamformer. The demodulated channel data are processed by parallel DAS and CF reduction paths to generate the weighted voxel response.

$$Q_m = -r_m \sin \phi_m + q_m \cos \phi_m, \quad (6)$$

where r_m and q_m are the real and imaginary parts of the delayed sample for the m th receive channel, and ϕ_m is the corresponding phase reference.

The demodulated channel data are coherently accumulated through the DAS path:

$$I_{\text{sum}} = \sum_{m=1}^M I_m, \quad Q_{\text{sum}} = \sum_{m=1}^M Q_m, \quad (7)$$

where M is the number of receive channels. The beamformed magnitude is computed as

$$B = \sqrt{I_{\text{sum}}^2 + Q_{\text{sum}}^2}. \quad (8)$$

To improve contrast and suppress incoherent responses under low-SNR air-coupled conditions, the beamformer also evaluates the coherence factor:

$$\text{CF} = \frac{I_{\text{sum}}^2 + Q_{\text{sum}}^2}{\sum_{m=1}^M (I_m^2 + Q_m^2)}. \quad (9)$$

The final voxel value is obtained as

$$V = \text{CF} \cdot B. \quad (10)$$

The DAS and CF computations are implemented as parallel pipelined reduction paths. This organization allows the beamformer to process the voxel stream continuously without introducing a separate post-processing stage.

D. Depth-Sparse SA Accumulation

Fig. 4 shows the proposed depth-sparse SA accumulation architecture. In conventional 3-D SA reconstruction, partial images from multiple transmit events must be stored and accumulated over the full voxel volume. This dense buffering becomes the dominant memory bottleneck for fully on-chip FPGA implementation. To reduce this overhead, the proposed design retains only the significant depth responses along each scanline.

The sparse memory format is shown in Fig. 5. Each scanline contains a header and a payload. The header includes a 9-bit min_coord field and a 54-bit valid mask, while the payload stores up to K selected voxel values. The min_coord field defines the base depth coordinate of the sparse window, and the valid mask indicates the relative positions of active samples within the window. During reconstruction, the valid mask and

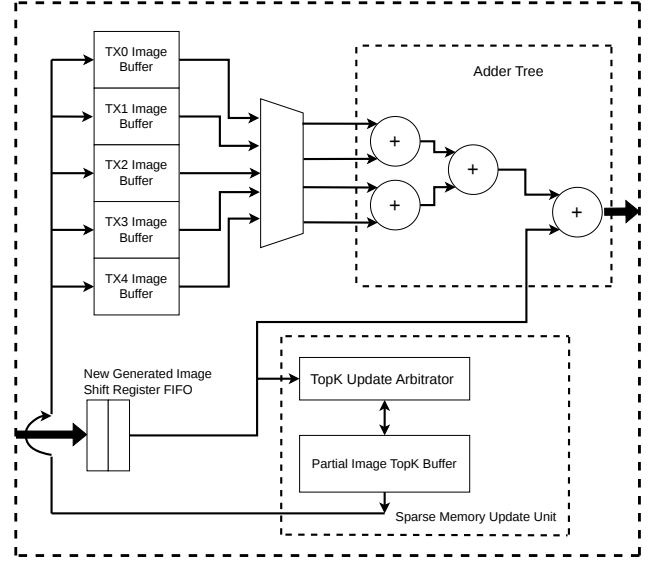


Fig. 4. Proposed depth-sparse SA accumulation architecture with multi-transmit image buffers, pipelined adder tree, and sparse memory update unit.

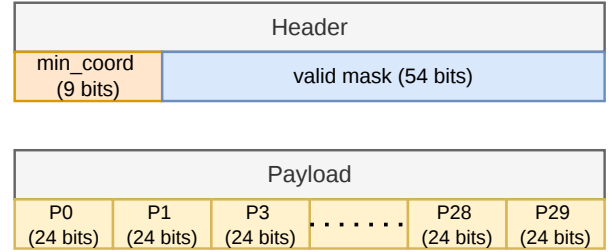


Fig. 5. Bit-level definition of the proposed sparse data structure. Each scanline is represented by a compact header and a fixed-depth payload.

the read pointer are used to recover the selected depth samples and add them into the final SA frame.

For a dense representation, each scanline requires $N_z B_v$ bits. In contrast, the proposed sparse format requires

$$9 + 54 + K B_v \quad (11)$$

bits, including the base coordinate, valid mask, and payload values. Therefore, the compression ratio is

$$R_{\text{comp}} = \frac{N_z B_v}{9 + 54 + K B_v}. \quad (12)$$

With $N_z = 284$, $B_v = 24$, and $K = 30$, the storage per scanline is reduced from 6816 bits to 783 bits, corresponding to an $8.70\times$ compression ratio and an 88.51% memory reduction.

The sparse buffer is updated through a streaming Top-K policy. Incoming voxel values are first checked by a magnitude threshold. Valid samples are inserted only when their coordinates fall within the current sparse window defined by min_coord . If the buffer is full, the update unit compares the incoming magnitude with the stored entries and replaces the least significant response when appropriate. This update policy preserves the dominant depth responses while keeping the payload size fixed. During frame addition, the reconstruction

logic scans the valid mask, reads the corresponding payload entries, and accumulates the selected samples into the output SA image. As a result, multi-transmit SA accumulation can be performed entirely on chip without storing dense intermediate volumes.

IV. EVALUATION

This section evaluates the proposed architecture from two perspectives: imaging performance and system-level comparison. We first describe the experimental setup, then analyze the imaging quality of the proposed design, and finally compare the overall system performance with prior work.

A. Experimental Setup

The proposed 3-D synthetic aperture ultrasonic imaging system was evaluated in both imaging simulation and hardware implementation. The imaging configuration used a 40-kHz air-coupled ultrasound system with 64 receive channels and 5 transmit events. The imaging range was set from 0.3 to 2.0 m with a field of view of $\pm 60^\circ$. A total of 69×69 scanlines and 284 depth samples were used to form the 3-D imaging volume. Point-target responses were generated using Field II for both on-axis and off-axis targets at different depths to evaluate spatial resolution and sidelobe behavior. For hardware evaluation, the proposed architecture was implemented in Vitis HLS and synthesized on the AMD/Xilinx ZCU102 FPGA platform. The overall system performance was evaluated in terms of imaging quality, hardware cost, and real-time throughput.

B. Sparse Memory Compression Analysis

Table II summarizes the memory reduction achieved by the proposed depth-sparse representation. The dense baseline stores all depth samples along each scanline, whereas the proposed format stores a 63-bit header and K selected voxel values. For the implemented configuration, each scanline contains $N_z = 284$ depth samples and each voxel is represented using 24 bits.

With $K = 30$, the proposed sparse representation retains 10.56% of the depth samples along each scanline. The per-scanline storage is reduced from 6816 bits to 783 bits, corresponding to an $8.70\times$ compression ratio and an 88.51% memory reduction. Across the full 69×69 angular grid, the storage requirement is reduced from 32.46 Mbits to 3.73 Mbits. This reduction is critical for keeping the synthetic-aperture accumulation buffers fully on chip on the ZCU102 FPGA platform. Since point-target responses in each scanline are strongly concentrated around the dominant echo depth after beamforming, the low-magnitude background samples contribute little to the final angular response. The K-sensitivity results in Section IV-D further verify that retaining only the dominant sparse payload preserves the measured beamwidth and PSL for the evaluated case.

C. Imaging Performance

Fig. 6 shows the normalized angular responses for on-axis and off-axis point targets at depths of 0.5, 1.0, and 1.5 m. The

TABLE II

MEMORY COMPRESSION ANALYSIS OF THE PROPOSED DEPTH-SPARSE REPRESENTATION.

Metric	Dense	Sparse ($K = 30$)
Depth samples per scanline	284	30
Retained voxel percentage	100%	10.56%
Header size per scanline	0 bits	63 bits
Payload size per scanline	6816 bits	720 bits
Total storage per scanline	6816 bits	783 bits
Compression ratio	$1.00\times$	$8.70\times$
Memory reduction	—	88.51%
Storage for 69×69 scanlines	32.46 Mbits	3.73 Mbits

colored curves represent the five individual transmit responses, while the black curve denotes their synthetic-aperture summation. The -6 dB beamwidth and peak sidelobe level (PSL) are used to evaluate angular resolution and sidelobe suppression, respectively. The PSL is defined as

$$\text{PSL} = \max_{\theta \in \Omega_{\text{SL}}} B_{\text{norm}}(\theta), \quad (13)$$

where $B_{\text{norm}}(\theta)$ is the normalized angular response and Ω_{SL} denotes the sidelobe region outside the mainlobe exclusion window.

For the on-axis target, the summed response consistently exhibits a narrower mainlobe than the individual transmit responses. At 0.5 m, the individual beamwidths are approximately 3.0° – 3.6° , whereas synthetic-aperture summation reduces the beamwidth to about 2.3° . At 1.0 m and 1.5 m, the summed beamwidth remains close to 2.5° , while the individual responses broaden to about 3.4° – 4.9° .

For the off-axis target at approximately $(30^\circ, 30^\circ)$, the mainlobe becomes slightly broader due to reduced aperture coherence under steering. Nevertheless, the summed response still provides a clear and well-localized peak at the expected angle. The synthetic-aperture beamwidths are approximately 2.68° , 2.90° , and 3.00° at depths of 0.5 m, 1.0 m, and 1.5 m, respectively, with corresponding PSL values of about -32.2 dB, -32.4 dB, and -31.7 dB.

Overall, these results show that the proposed synthetic-aperture beamforming improves angular resolution, suppresses sidelobe artifacts, and maintains robust focusing across different target positions and depths.

D. K-Sensitivity Analysis

To evaluate the selected sparse payload depth, a K-sensitivity study was performed using an on-axis point target at 1 m. The payload depth was varied from $K = 20$ to $K = 40$, and the summed synthetic-aperture responses were compared using the -6 dB beamwidth and PSL.

As shown in Fig. 7, the responses for $K = 20$, $K = 30$, and $K = 40$ almost completely overlap. All cases achieve the same -6 dB beamwidth of 2.56° and a PSL of approximately -37.2 dB, indicating that the dominant point-target response is preserved. Since increasing K beyond 30 provides no observable improvement in this case but increases memory usage, $K = 30$ is selected for implementation.



Fig. 6. Normalized angular responses for on-axis and off-axis point targets at different depths. The synthetic aperture summation achieves a narrower and well-focused mainlobe compared with individual transmit responses.

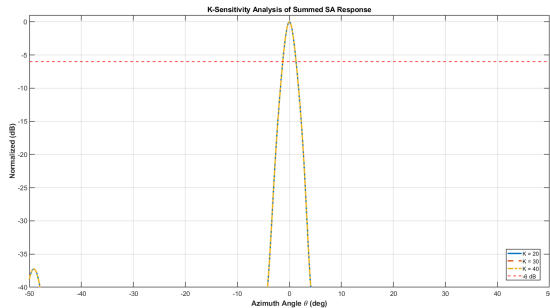


Fig. 7. K-sensitivity analysis of the summed synthetic-aperture response for an on-axis point target at 1.0 m. The nearly overlapped curves show that the point-target response is preserved for $K = 20$ –40.

E. Hardware Resource Utilization and Scalability

In addition to imaging quality and throughput, hardware cost is an important metric for evaluating the practicality of the proposed FPGA-based architecture. Table III summarizes the post-implementation resource utilization of the proposed system operating at 125 MHz.

The implementation occupies 51,757 LUTs, 37,434 FFs,

1,187 DSPs, and 1,528 BRAMs. The LUT and FF utilization remain moderate, at 18.88% and 6.83%, respectively, while the DSP utilization reaches 47.10%. In contrast, the BRAM/FIFO utilization reaches 83.77%, indicating that the proposed architecture is primarily memory-bound rather than logic-bound.

This memory demand mainly comes from the fully on-chip reconstruction constraint, where sparse voxel buffers are used to store intermediate image responses without relying on external memory access. The sparse payload depth is therefore selected as $K = 30$ to fit within the BRAM budget of the target ZCU102 platform. Increasing K to 35 would nearly exhaust the available BRAM resources.

Overall, the current implementation represents a resource-limited design point that sustains real-time operation while remaining fully implementable on the target FPGA. Further throughput scaling would require additional processing lanes, deeper sparse buffers, a larger FPGA device, or external memory support.

F. Comparison with Prior Work

Table IV summarizes the system configurations of the proposed design and prior 3-D ultrasonic imaging systems.

TABLE III

POST-IMPLEMENTATION HARDWARE RESOURCE UTILIZATION OF THE PROPOSED FPGA-BASED IMAGING SYSTEM.

Resource	Utilization	Percentage
LUT	51,757	18.88%
FF	37,434	6.83%
DSP	1,187	47.10%
BRAM/FIFO	1,528	83.77%
Operating frequency	125 MHz	

As shown in the table, the compared systems differ in array geometry, transmit strategy, field of view, voxel configuration, sensing range, and hardware platform. Therefore, the performance comparison in Table V is intended as a system-level reference only and should not be interpreted as a direct benchmark across identical hardware and imaging conditions.

Table V further compares the reported spatial resolution and frame rate of related 3-D air-coupled ultrasonic imaging systems. Although the system configurations are different, the proposed design demonstrates competitive angular resolution and frame-rate performance among the compared works.

Compared with [6], the proposed system provides substantially finer angular resolution. This improvement mainly comes from per-voxel delay compensation, synthetic aperture accumulation over multiple transmit events, and coherence factor weighting for sidelobe suppression. In contrast, [6] focuses on real-time volumetric imaging with hemispherical-pulse and dynamic-transmit-beamforming modes, but does not employ the same per-voxel SA reconstruction and coherence-based weighting strategy.

Compared with [7], the proposed design achieves comparable spatial resolution while delivering a higher frame rate on an FPGA platform. While [7] improves image quality through a one-shot transmitter and ASIC-based per-voxel RX beamfocusing to increase effective transmit energy and SNR, this work enhances focusing through synthetic aperture accumulation and coherence factor weighting. This reflects a different hardware–algorithm tradeoff.

The high throughput of the proposed system is further enabled by its fully on-chip FPGA architecture. By avoiding external memory access for intermediate image formation, the design reduces bandwidth and transfer overhead, which is critical for efficient synthetic aperture reconstruction. Overall, the proposed architecture effectively combines image-quality enhancement with practical real-time FPGA implementation.

V. CONCLUSION

This paper has presented a depth-sparse 3-D synthetic aperture ultrasonic imaging architecture for real-time air-coupled environment sensing. The proposed design addresses the two major challenges of 3-D synthetic aperture imaging on FPGA: the high complexity of voxel-wise delay generation and the large memory demand of multi-transmit image accumulation. To overcome these issues, a hardware-oriented beamforming architecture was developed by combining hybrid delay generation, streaming synthetic aperture beamforming, and depth-sparse on-chip accumulation.

The experimental results demonstrate that the proposed architecture achieves improved angular resolution through synthetic aperture summation and maintains robust imaging performance across different target positions and depths. In particular, the system attains a spatial resolution of 2.3° while sustaining a frame rate of 68.56 frames/s on the ZCU102 FPGA platform. By adopting a fully on-chip reconstruction flow, the proposed design avoids external memory access during intermediate image formation and translates algorithmic image-quality enhancement into a practical real-time implementation.

These results confirm that high-resolution 3-D air-coupled ultrasonic imaging can be effectively realized on FPGA through joint algorithm–architecture co-design. The proposed system therefore provides a practical solution for real-time ultrasonic environment sensing in applications such as drones, smart factories, and warehouse automation.

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TABLE IV

SYSTEM CONFIGURATION COMPARISON OF THE PROPOSED SYSTEM AND PRIOR 3-D ULTRASONIC IMAGING SYSTEMS.

Parameter	This Work	JSSC'22 [7]	IEEE Trans. [6]
Array type	2-D sunflower sparse array	8 × 8 bulk piezo array	8 × 8 uniform rectangular array
Number of RX channels	64	64	64
Transmit scheme	5-TX synthetic aperture	one-shot TX, 2 × 2 TX enabled	HP: single TX; DTB also supported
Center frequency	40 kHz	40 kHz	40 kHz
Field of view	±60°	±30°	±50°
Angular grid	69 × 69	64 × 64	51 × 51
Depth samples	284	100	3500
Total voxel number	69 × 69 × 284	64 × 64 × 100	51 × 51 × 3500
Processing platform	FPGA (ZCU102)	ASIC (180 nm) + FPGA + ESP32	FPGA SoC + PC/GPU
On-chip reconstruction	Yes	Yes	No

TABLE V

PERFORMANCE COMPARISON WITH PRIOR 3-D ULTRASONIC IMAGING SYSTEMS.

Metric	This Work	JSSC'22 [7]	IEEE Trans. [6]
Best Spatial Resolution (°)	2.3	2.7	HP: 14, DTB: 12
Frame Rate (fps)	68.56	24	HP: 29, DTB: 0.015
Voxel Throughput (Mvox/s)	92.7	9.83	HP: 264.0 DTB: 0.137

[13] W. Wang and Z. Feng, "Portable and cost-effective handheld ultrasound system utilizing FPGA-based synthetic aperture imaging," *IEEE Open Journal of Nanotechnology*, vol. 5, pp. 107–115, 2024.

[14] X. Zhang, S. Qi, Y. Li, L. Xu, and J. Ma, "Ultrafast ultrasound imaging system with beamforming on a single FPGA," *IEEE Transactions on Instrumentation and Measurement*, vol. 74, pp. 1–9, 2025.



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systems.

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